🛡️ Task 2: Network Security & Scanning Report

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Internship ID: APSPL2631283

Timeline: Days 13–24

1. Executive Summary

The objective of this task was to master reconnaissance, vulnerability scanning, and network traffic analysis. This report documents the identification of services on a target machine, the discovery of a critical backdoor, the interception of plaintext credentials, and the implementation of defensive firewall rules.

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2. Reconnaissance & Service Enumeration

Following the "Information Gathering" phase, I performed active reconnaissance to map the target's attack surface.

Step 1: Port & Service Scanning (Nmap) I used Nmap to identify active services and their respective versions on the target IP.

- **Command:** `nmap -sV -O 192.168.56.102`
- **Findings:** Discovered Port 21 (FTP), Port 23 (Telnet), and Port 80 (HTTP) were open. The scan identified the service `vsFTPd 2.3.4` running on Port 21.

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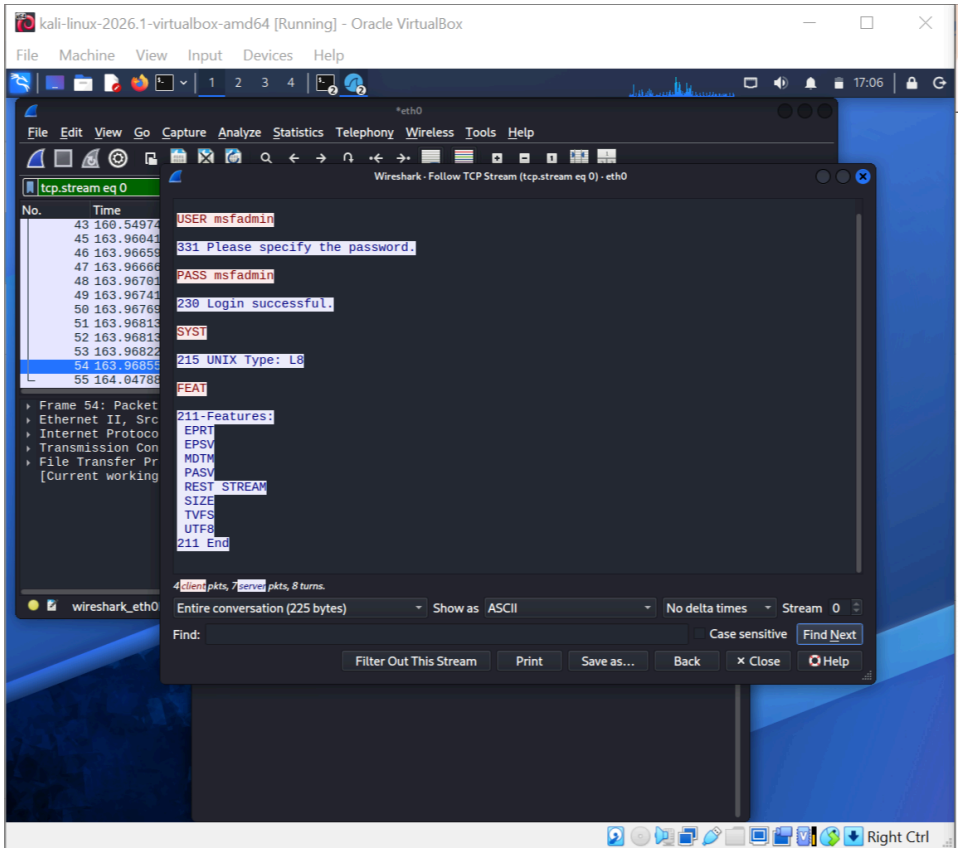
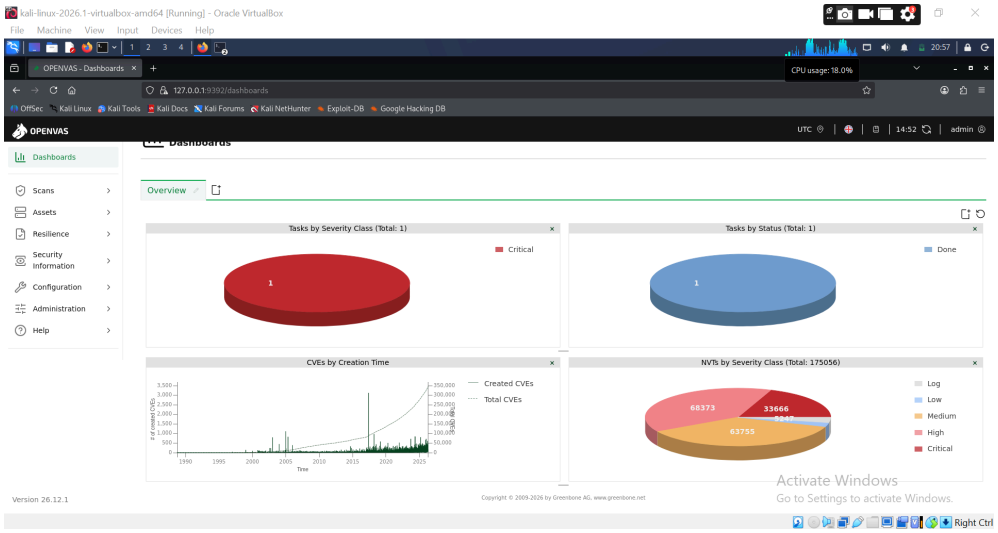
3. Vulnerability Scanning (The "Analysis" Pillar)

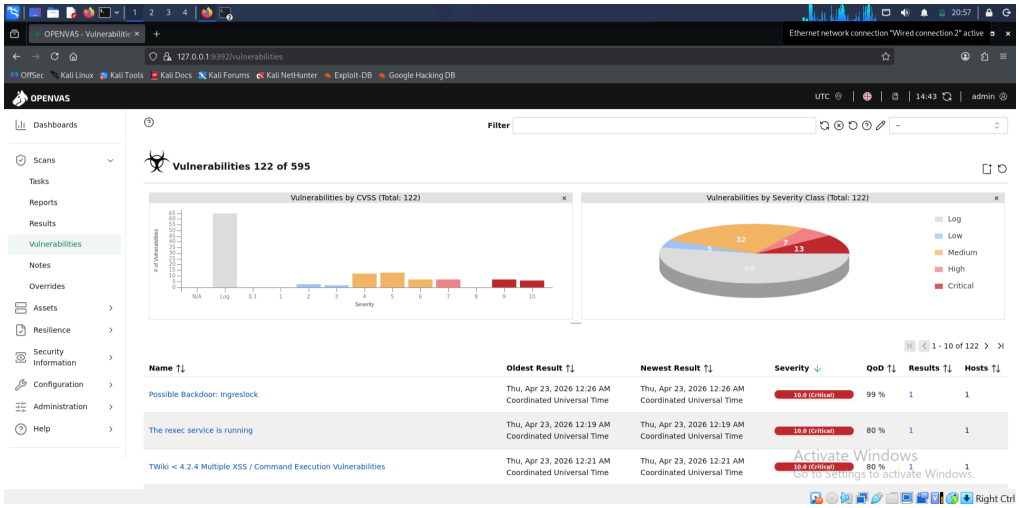
To move beyond basic port mapping, I conducted an automated vulnerability assessment to determine the risk level of the identified services.

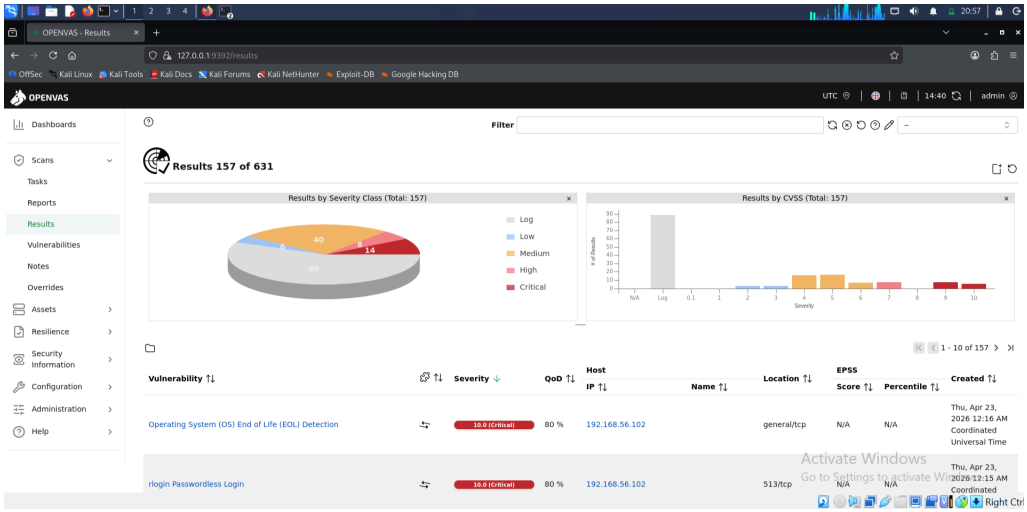
Step 1: GVM/OpenVAS Assessment I executed a vulnerability scan using the Greenbone Vulnerability Management suite.

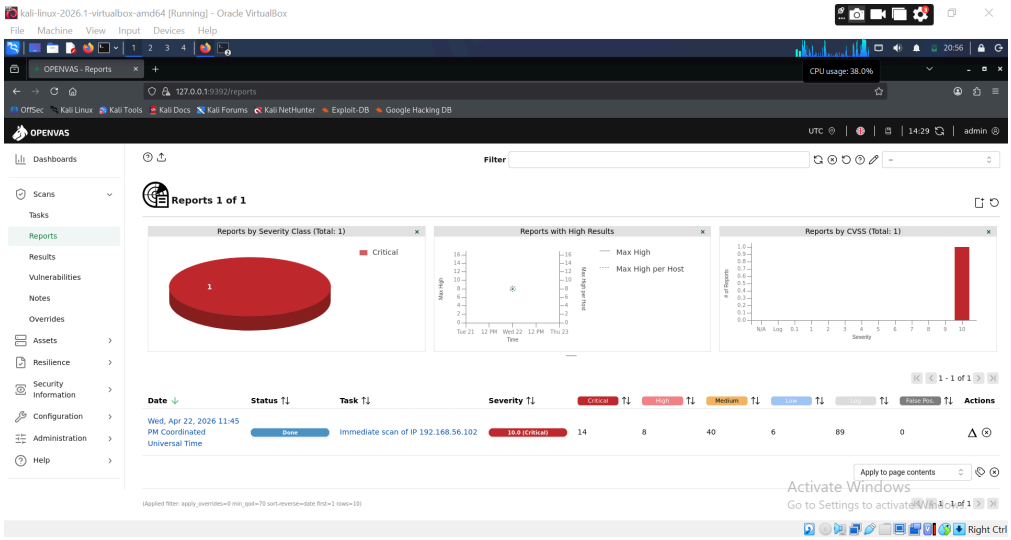
- **Result:** Identified a **CVSS 10.0 (Critical)** vulnerability.
- **Vulnerability:** vsFTPd 2.3.4 Backdoor Command Execution.
- **Impact:** This flaw allows an unauthenticated attacker to execute commands as root, leading to total system compromise.

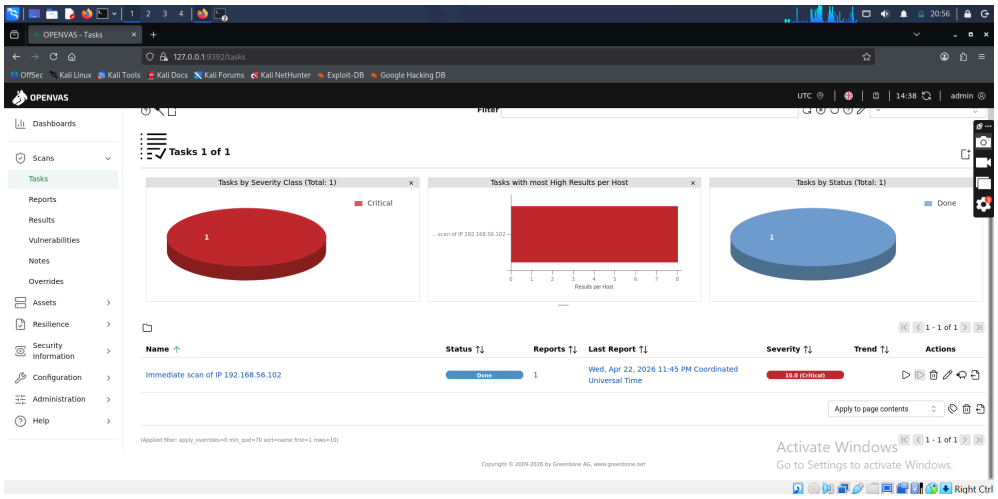

kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
*eth0
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.stream eq 0
Wireshark · Follow TCP Stream (tcp.stream eq 0) · eth0
USER msfadmin
331 Please specify the password.
PASS msfadmin
230 Login successful.
SYST
215 UNIX Type: L8
FEAT
211-Features:
EPRT
EPSV
MDTM
PASV
REST STREAM
SIZE
TVFS
UTF8
211 End
Entire conversation (225 bytes)
Show as ASCII
No delta times
Stream 0
Find:
Case sensitive
Find Next
Filter Out This Stream
Print
Save as...
Back
Close
Help
wireshark_eth0
Right Ctrl


kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
OPENVAS - Dashboards
127.0.0.1:9392/dashboards
OPENVAS
Dashboards
Scans
Assets
Resilience
Security Information
Configuration
Administration
Help
Overview
Tasks by Severity Class (Total: 1)
Critical
Tasks by Status (Total: 1)
Done
CVEs by Creation Time
Created CVEs
Total CVEs
NVTs by Severity Class (Total: 175056)
Log
Low
Medium
High
Critical
Version 26.12.1
Activate Windows
Go to Settings to activate Windows.
Right Ctrl


OPENVAS - Vulnerabilities
127.0.0.1:9392/vulnerabilities
OPENVAS
Dashboards
Scans
Tasks
Reports
Results
Vulnerabilities
Notes
Overrides
Assets
Resilience
Security Information
Configuration
Administration
Help
Filter
Vulnerabilities 122 of 595
Vulnerabilities by CVSS (Total: 122)
Vulnerabilities by Severity Class (Total: 122)
Log
Low
Medium
High
Critical
1 - 10 of 122
Name
Oldest Result
Newest Result
Severity
QoD
Results
Hosts
Possible Backdoor: Ingreslock
Thu, Apr 23, 2026 12:26 AM Coordinated Universal Time
10.0 (Critical)
99 %
The rexec service is running
Thu, Apr 23, 2026 12:19 AM Coordinated Universal Time
80 %
TWiki < 4.2.4 Multiple XSS / Command Execution Vulnerabilities
Thu, Apr 23, 2026 12:21 AM Coordinated Universal Time
Activate Windows
Go to Settings to activate Windows.
Right Ctrl


OPENVAS - Results
127.0.0.1:9392/results
OPENVAS
Dashboards
Scans
Tasks
Reports
Results
Vulnerabilities
Notes
Overrides
Assets
Resilience
Security Information
Configuration
Administration
Help
Filter
Results 157 of 631
Results by Severity Class (Total: 157)
Log
Low
Medium
High
Critical
Results by CVSS (Total: 157)
1 - 10 of 157
Vulnerability
Severity
QoD
Host
IP
Name
Location
EPSS
Score
Percentile
Created
Operating System (OS) End of Life (EOL) Detection
10.0 (Critical)
80 %
192.168.56.102
general/tcp
N/A
Thu, Apr 23, 2026 12:16 AM Coordinated Universal Time
rlogin Passwordless Login
513/tcp
Thu, Apr 23, 2026 12:15 AM Coordinated
Activate Windows
Go to Settings to activate Windows.
Right Ctrl


kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
OPENVAS - Reports
CPU usage: 38.0%
127.0.0.1:9392/reports
OPENVAS
Dashboards
Scans
Tasks
Reports
Results
Vulnerabilities
Notes
Overrides
Assets
Resilience
Security Information
Configuration
Administration
Help
Filter
Reports 1 of 1
Reports by Severity Class (Total: 1)
Critical
Reports with High Results
Max High
Max High per Host
Reports by CVSS (Total: 1)
1 - 1 of 1
Date
Status
Task
Severity
Critical
High
Medium
Low
Log
False Pos.
Actions
Wed, Apr 22, 2026 11:45 PM Coordinated Universal Time
Done
Immediate scan of IP 192.168.56.102
10.0 (Critical)
14
8
40
6
89
0
Apply to page contents
Activate Windows
Go to Settings to activate Windows.
Right Ctrl



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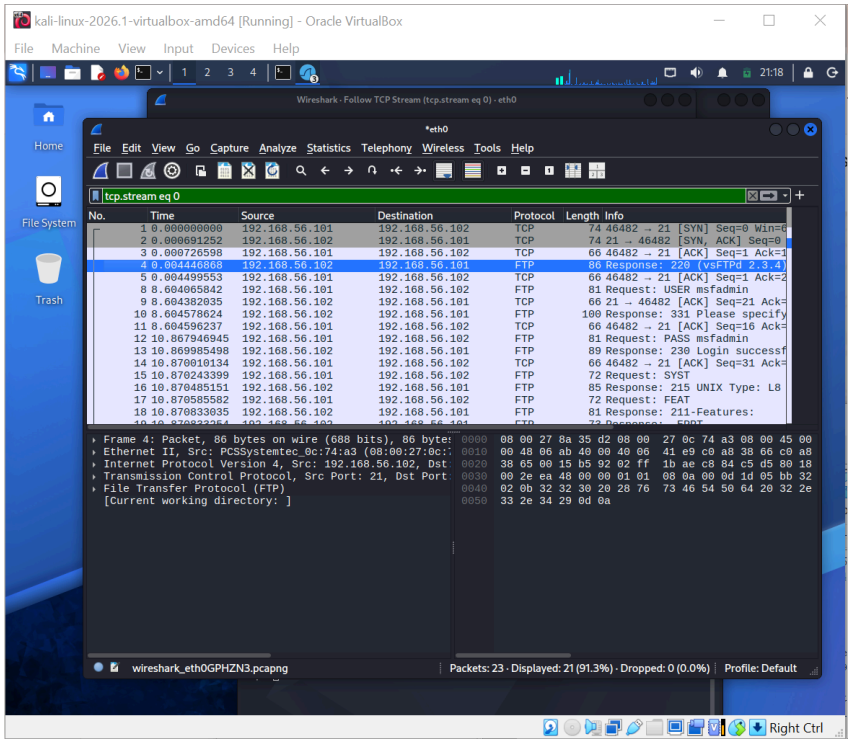
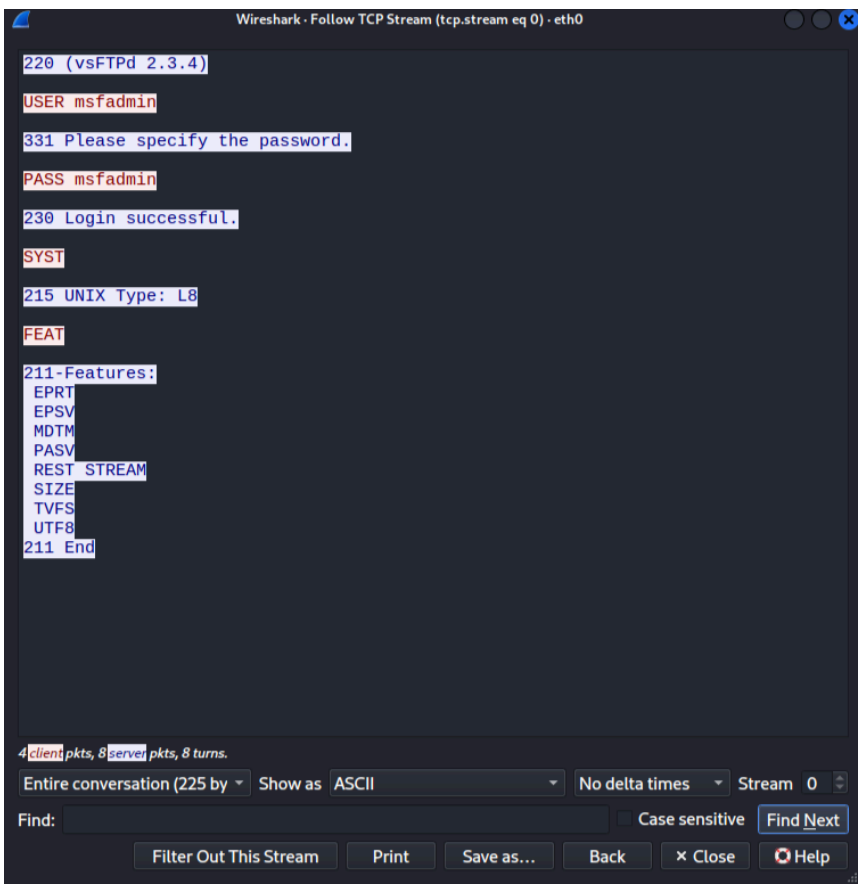
4. Network Traffic Analysis (Wireshark)

I used Wireshark to perform deep packet inspection (DPI) on the `eth0` interface to observe how data moves between the attacker and target.

Step 1: Credential Sniffing By following the **TCP Stream** of an FTP session, I demonstrated the insecurity of unencrypted protocols.

- **Data Captured:** Cleartext username (`msfadmin`) and password (`msfadmin`).
- **Conclusion:** Verified that protocols lacking TLS/SSL encryption are susceptible to sniffing and MITM attacks.


kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Wireshark · Follow TCP Stream (tcp.stream eq 0) · eth0
*eth0
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.stream eq 0
No. Time Source Destination Protocol Length Info
1 0.000000000 192.168.56.101 192.168.56.102 TCP 74 46482 → 21 [SYN] Seq=0 Win=6
2 0.000691252 192.168.56.102 192.168.56.101 TCP 74 21 → 46482 [SYN, ACK] Seq=0
3 0.000726598 192.168.56.101 192.168.56.102 TCP 66 46482 → 21 [ACK] Seq=1 Ack=1
4 0.004446868 192.168.56.102 192.168.56.101 FTP 86 Response: 220 (vsFTPd 2.3.4)
5 0.004499553 192.168.56.101 192.168.56.102 TCP 66 46482 → 21 [ACK] Seq=1 Ack=2
8 8.604065842 192.168.56.101 192.168.56.102 FTP 81 Request: USER msfadmin
9 8.604382035 192.168.56.102 192.168.56.101 TCP 66 21 → 46482 [ACK] Seq=21 Ack=
10 8.604578624 192.168.56.102 192.168.56.101 FTP 100 Response: 331 Please specify
11 8.604596237 192.168.56.101 192.168.56.102 TCP 66 46482 → 21 [ACK] Seq=16 Ack=
12 10.867946945 192.168.56.101 192.168.56.102 FTP 81 Request: PASS msfadmin
13 10.869985498 192.168.56.102 192.168.56.101 FTP 89 Response: 230 Login successf
14 10.870010134 192.168.56.101 192.168.56.102 TCP 66 46482 → 21 [ACK] Seq=31 Ack=
15 10.870243399 192.168.56.101 192.168.56.102 FTP 72 Request: SYST
16 10.870485151 192.168.56.102 192.168.56.101 FTP 85 Response: 215 UNIX Type: L8
17 10.870585582 192.168.56.101 192.168.56.102 FTP 72 Request: FEAT
18 10.870833035 192.168.56.102 192.168.56.101 FTP 81 Response: 211-Features:
Frame 4: Packet, 86 bytes on wire (688 bits), 86 byte
Ethernet II, Src: PCSSystemtec_0c:74:a3 (08:00:27:0c:
Internet Protocol Version 4, Src: 192.168.56.102, Dst
Transmission Control Protocol, Src Port: 21, Dst Port
File Transfer Protocol (FTP)
[Current working directory:]
0000 08 00 27 8a 35 d2 08 00 27 0c 74 a3 08 00 45 00
0010 00 48 06 ab 40 00 40 06 41 e9 c0 a8 38 66 c0 a8
0020 38 65 00 15 b5 92 02 ff 1b ae c8 84 c5 d5 80 18
0030 00 2e ea 48 00 00 01 01 08 0a 00 0d 1d 05 bb 32
0040 02 0b 32 32 30 20 28 76 73 46 54 50 64 20 32 2e
0050 33 2e 34 29 0d 0a
wireshark_eth0GPHZN3.pcapng
Packets: 23 · Displayed: 21 (91.3%) · Dropped: 0 (0.0%)
Profile: Default
Right Ctrl


Wireshark · Follow TCP Stream (tcp.stream eq 0) · eth0
220 (vsFTPd 2.3.4)
USER msfadmin
331 Please specify the password.
PASS msfadmin
230 Login successful.
SYST
215 UNIX Type: L8
FEAT
211-Features:
EPRT
EPSV
MDTM
PASV
REST STREAM
SIZE
TVFS
UTF8
211 End
4 client pkts, 8 server pkts, 8 turns.
Entire conversation (225 by
Show as ASCII
No delta times
Stream 0
Find:
Case sensitive
Find Next
Filter Out This Stream
Print
Save as…
Back
Close
Help

5. Firewall Basics (Defensive Configuration)

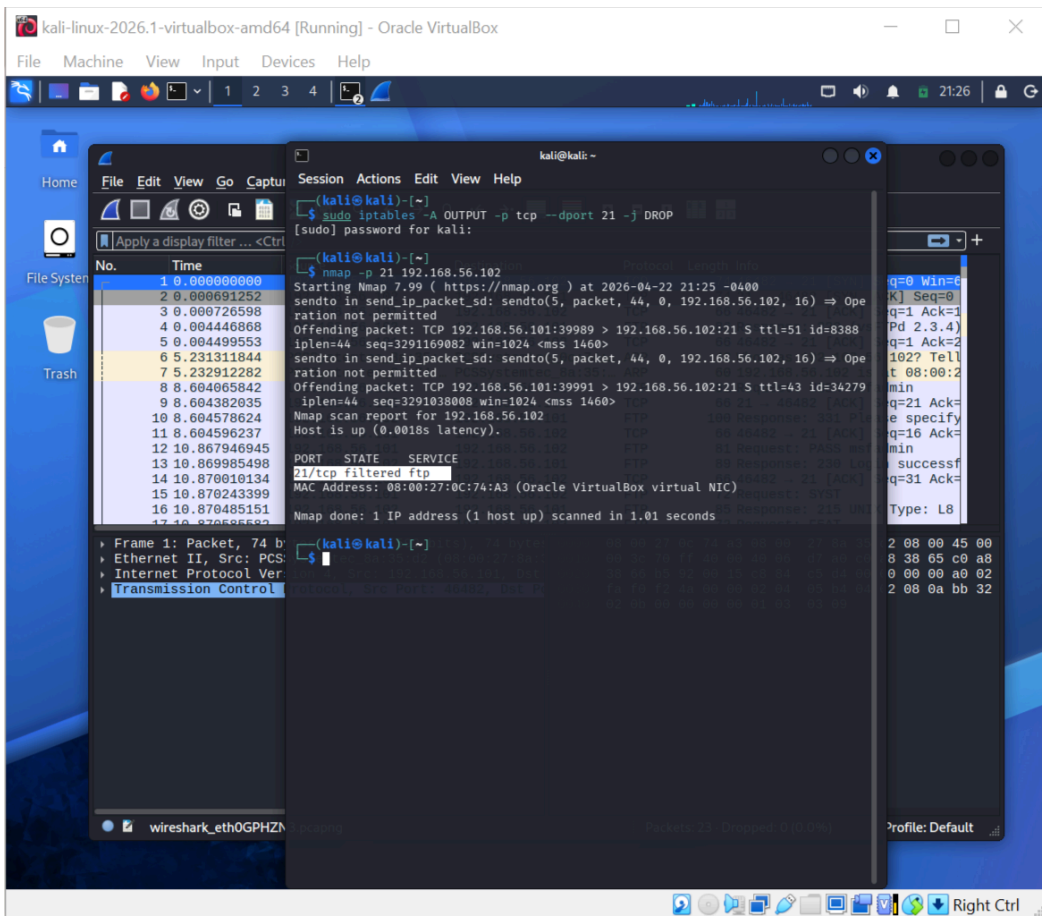
Using the `iptables` utility, I configured host-based rules to demonstrate the "Deny" security posture.

Step 1: Blocking Traffic I created a rule to deny all outgoing traffic to the vulnerable FTP port.

- **Command:** `sudo iptables -A OUTPUT -p tcp --dport 21 -j DROP`

Step 2: Verification of Block I re-ran the Nmap scan to verify the firewall's effectiveness.

- **Nmap Result:** The state of Port 21 changed from **"Open"** to **"Filtered,"** proving the firewall successfully dropped the connection attempt.



6. Conclusion

The reconnaissance and scanning phase for Days 13–24 is complete. All deliverables, including the Nmap scan report, OpenVAS vulnerability report, and firewall verification, have been documented and uploaded to the GitHub repository.